

⇒ EM algorithm

* Basic set up: (1) observed data: Y

$$(2) L(\theta, Y) = P(Y|\theta)$$

$P(Y|\theta)$ involves an integral
or sum.

* Why do we need EM?

(1) Avoid the integral in optimization

(2) Borrow some analytical optimization
result,

Hidden (latent, missing) Data: Z

$$P(Y|\theta) = \int P(Y, Z|\theta) dz$$

* Basic terminologies:

(1) observed likelihood:

$$L_{\text{obs}}(\theta) = P(Y|\theta)$$

observed log likelihood:

$$l_{\text{obs}}(\theta) = \log P(Y|\theta)$$

(2) Complete likelihood: $L_{\text{comp}}(\theta) = P(Y, Z|\theta)$

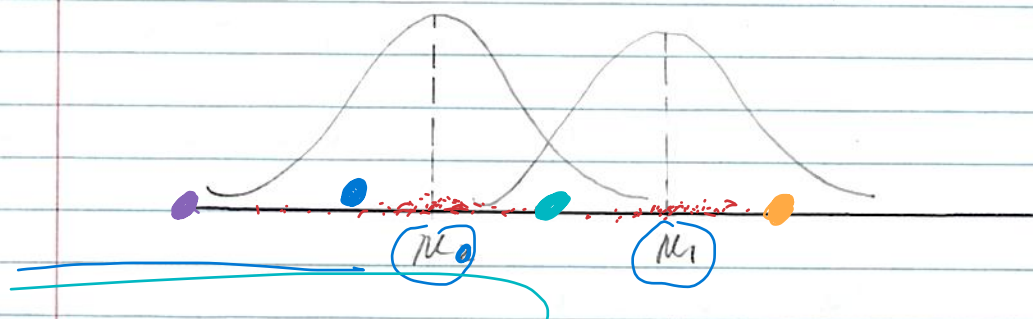
Complete log likelihood:

$$l_{\text{comp}}(\theta) = \log P(Y, Z|\theta)$$

Example 1: Fitting Mixture Models

=> example:

1) Mixture Model 1.



$$Y_i | Z_i = 1 \sim N(\mu_1, 1)$$

$$Y_i | Z_i = 0 \sim N(\mu_0, 1)$$

in here, $Z_i \sim \text{Bern}(P)$

observed Data: $Y_i \quad i=1, 2, \dots, n$

Hidden Data: $Z_i \quad i=1, \dots, n$

Observed likelihood $\theta = (\mu_0, \mu_1, P)$

$$P(y_1, \dots, y_n | \theta)$$

$$\Delta = \prod_{i=1}^n \left[\varphi(y_i | \mu_0, 1)^{1-P} + \varphi(y_i | \mu_1, 1)^P \right]$$

Completed likelihood:

$$P(\underline{y_1, \dots, y_n}, \underline{z_1, \dots, z_n} | \theta)$$

$$= \prod_{i=1}^n P(y_i, z_i | \theta)$$

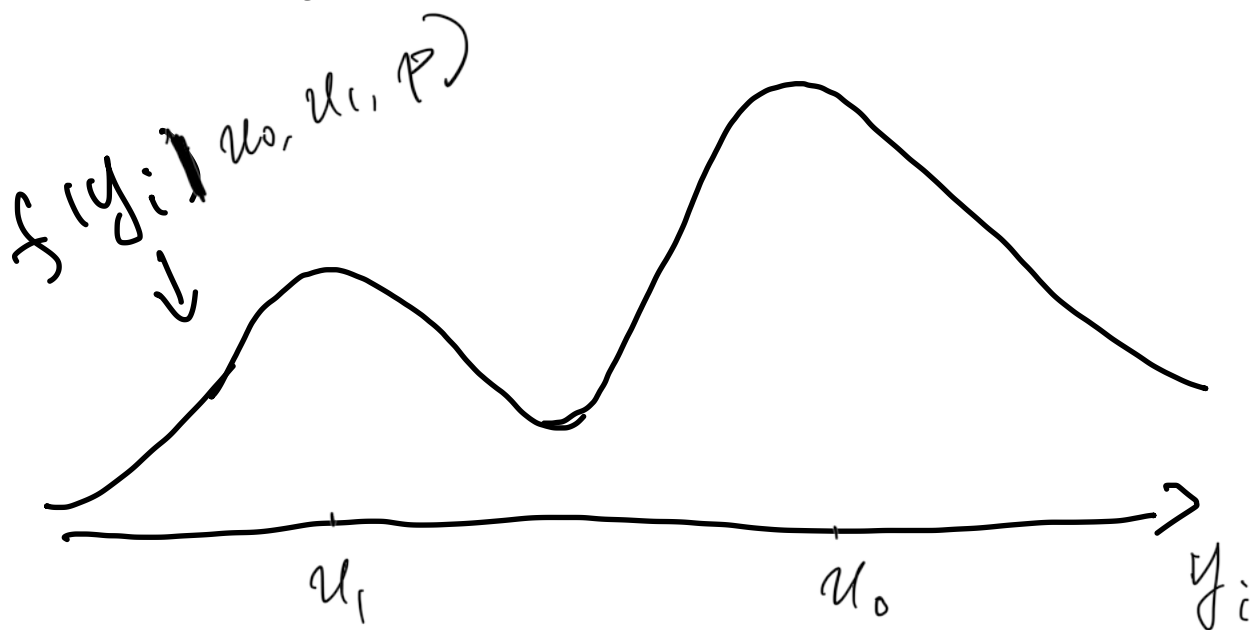
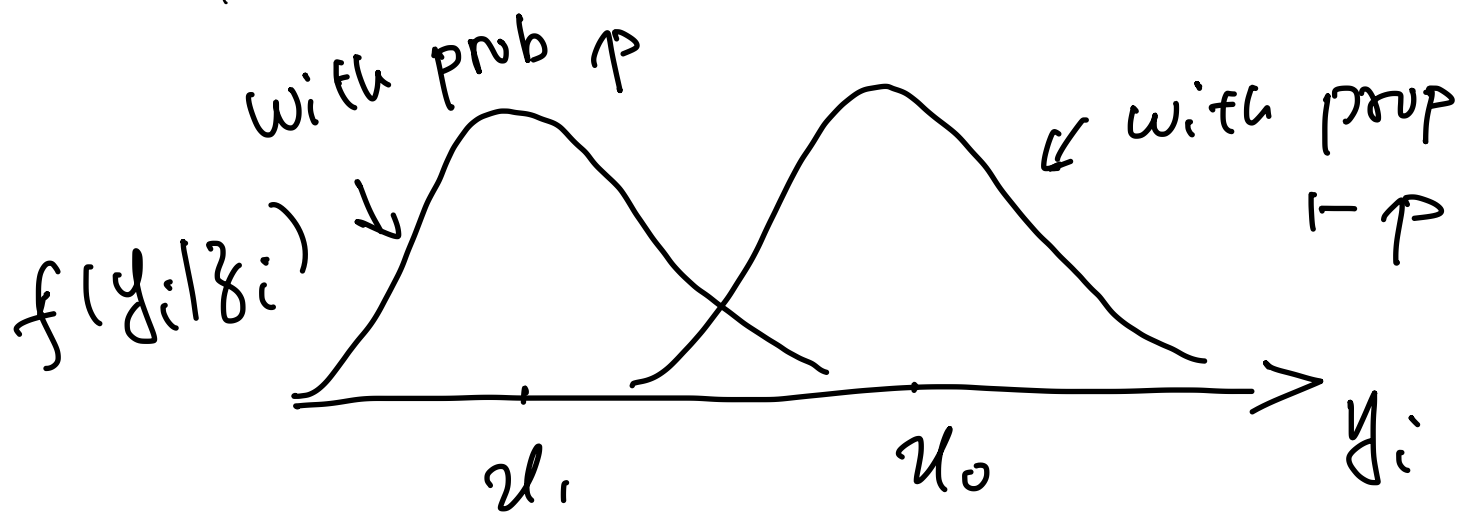
$$= \prod_{i=1}^n P(y_i | \theta, z_i) \cdot P(z_i | \theta)$$

$$= \prod_{i=1}^n \varphi(y_i | \mu_{z_i}, 1) p^{z_i} (1-p)^{1-z_i}$$

observed likelihood:

$$\prod_{i=1}^n \sum_{z_i=0}^1 \varphi(y_i | \mu_{z_i}, 1) \underbrace{p^{z_i} (1-p)^{1-z_i}}_{p(y_i | z_i) \cdot p(z_i)}$$

$$= \prod_{i=1}^n [\varphi(y_i | \mu_1, 1) \cdot p + \varphi(y_i | \mu_0, 1) \cdot (1-p)]$$



$\Rightarrow$ EM

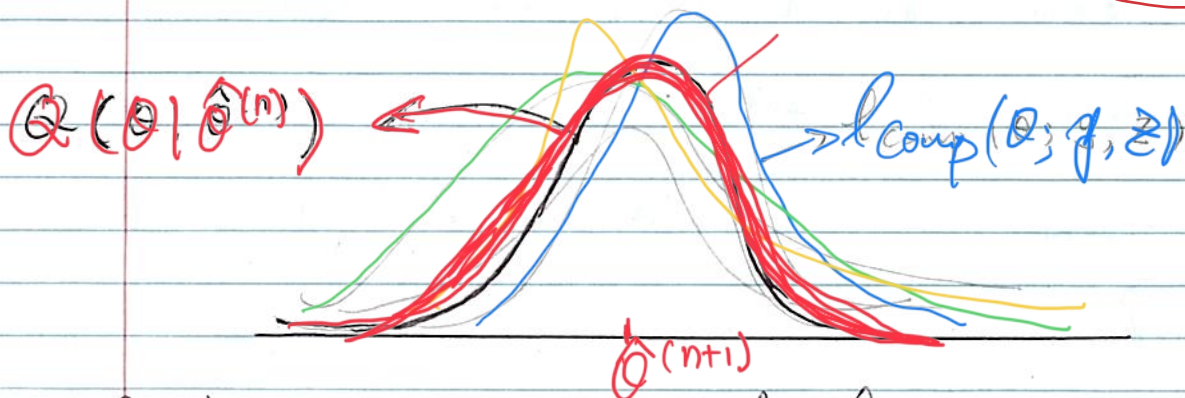
$\Rightarrow$ starting from $\hat{\theta}^{(0)}$, iterate
the E-step, and M-step
For $n=0, 1, 2, \dots$

E-step : approximate

$\ell_{\text{obs}}(\theta)$ with

$$Q(\theta | \hat{\theta}^{(n)}) = E_{z | \hat{\theta}^{(n)}, y} [\ell_{\text{comp}}(\theta, y, z)]$$

Note that: Q is not $\ell_{\text{comp}}(\theta; y, E(z | \hat{\theta}^{(n)}))$



Q is an average of $\ell_{\text{comp}}(\theta, y, z)$ with respect to $p(z | \theta^{(n)}, y)$

M-step

$$\hat{\theta}^{(n+1)} = \arg\max_{\theta} Q(\theta | \hat{\theta}^{(n)})$$

⇒ example 1 (for mixture)

observed data: $y_1, \dots, y_m$

comp ($\theta: y_i, z_i$)

$$= \log \left(\prod_{i=1}^m p^{z_i} (1-p)^{1-z_i} \psi(y_i | \mu_{z_i}, 1) \right)$$

$$= \sum_{i=1}^m \left[z_i \log p + (1-z_i) \log(1-p) + \log \psi(y_i | \mu_{z_i}, 1) \right]$$

Note that $y_i | z_i \sim N(\mu_{z_i}, 1)$

$$\psi(y_i | \mu_{z_i}, 1) = \frac{1}{\sqrt{2\pi}} \exp \left\{ -\frac{1}{2} \left(\frac{y_i - \mu_{z_i}}{\sigma} \right)^2 \right\}$$

$$= \frac{1}{\sqrt{2\pi}} \exp \left\{ -\frac{(y_i - \mu_{z_i})^2}{2} \right\}$$

$$\log \psi(y_i | \mu_{z_i}, 1)$$

$$= \log \frac{1}{\sqrt{2\pi}} + \log \exp \left(-\frac{(y_i - \mu_{z_i})^2}{2} \right)$$

$$= -\frac{(y_i - \mu_{z_i})^2}{2} - \log \sqrt{2\pi}$$

$$= \sum_{i=1}^m z_i (\log p - \log(1-p)) + m \log(1-p)$$

$$- \frac{1}{2} \sum_{i=1}^m (y_i - \mu_{z_i})^2 + C \text{ (constant)}$$

$(z_i = 1, \text{ or } 0)$

$$= \sum_{i=1}^m \underbrace{z_i}_{z_i^{(n)}} (\log p - \log(1-p)) + m \log(1-p)$$

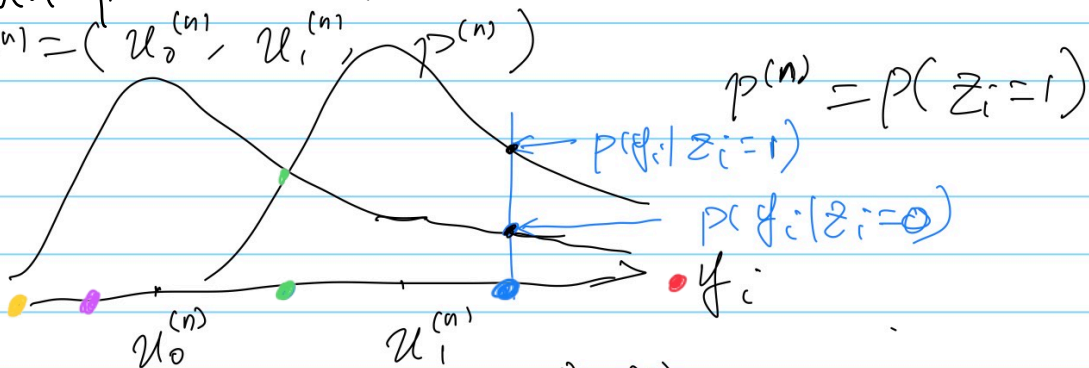
$$- \frac{1}{2} \sum_{i=1}^m (y_i - \mu_{z_i})^2$$

→ z_i need to be averaged away

E-Step: finding soft classification probabilities $p_i^{(n)}$ for fitting mixture distribution with EM (better regard n as t)

Given previous step

$$\theta^{(n)} = (\mu_0^{(n)}, \mu_1^{(n)}, p^{(n)})$$



$$p(z_i=1 | y_i, \theta^{(n)}) = \frac{p(y_i | z_i=1) p^{(n)}}{p(y_i | z_i=0) \cdot (1 - p^{(n)}) + p(y_i | z_i=1) \cdot p^{(n)}}$$

$$\equiv \hat{p}_i^{(n)} \text{ (Soft classification)}$$

	y_i	$\hat{p}_i^{(n)}$	$p(y_i z_i=0) (1 - p^{(n)})$	$p(y_i z_i=1) p^{(n)}$
1	x	$\frac{25}{45}$	20	25
2	x	0.2	x	x
3	x	0.9	,	,
.	x	.	.	.
.	x	.	.	.
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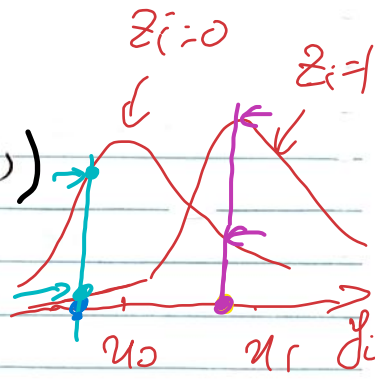
$$\hat{p}^{(n+1)} = \frac{\sum \hat{p}_i^{(n)}}{m}$$

E-step: Find $P(z_i = 1 | y_i, \hat{\theta}^{(n)})$

suppose we have found:

$$P(z_i = 1 | y_i; \hat{\theta}^{(n)}) = \hat{p}_i^{(n)}$$

$$P(z_i = 0 | y_i; \hat{\theta}^{(n)}) = 1 - \hat{p}_i^{(n)}$$



$$\begin{aligned} Q(\theta | \hat{\theta}^{(n)}) &= E_{z_i | \hat{\theta}^{(n)}, y_i} (l_{\text{comp}}(\theta, y, z)) \\ &= E_{z_i} \left[\sum_{i=1}^m z_i (\log p - \log(1-p)) + m \log(1-p) - \frac{1}{2} \sum_{i=1}^m (y_i - \mu_{z_i})^2 \right] \\ &= \sum_{i=1}^m E(z_i) \cdot \log \frac{p}{1-p} + m \cdot \log(1-p) - \frac{1}{2} \sum_{i=1}^m E[(y_i - \mu_{z_i})^2] \\ &= \sum_{i=1}^m \hat{p}_i^{(n)} \cdot \log \frac{p}{1-p} + m \cdot \log(1-p) \\ &\quad - \frac{1}{2} \sum_{i=1}^m [(y_i - \mu_0)^2 (1 - \hat{p}_i^{(n)}) + (y_i - \mu_1)^2 \hat{p}_i^{(n)}] \end{aligned}$$

M-step:

$$\begin{aligned} \hat{p}^{(n+1)} &= \arg \max_p \left[\sum_{i=1}^m \hat{p}_i^{(n)} \log \frac{p}{1-p} + m \log(1-p) \right] \\ &= \frac{\sum_{i=1}^m \hat{p}_i^{(n)}}{n} \end{aligned}$$

$Q^{(n)}(p)$

Differentiate Q with respect to p to find this optimizer for p

$$\begin{aligned} \frac{\partial Q^{(n)}(p)}{\partial p} &= \sum_{i=1}^m \hat{p}_i^{(n)} \cdot \left(\frac{1}{p} + \frac{1}{1-p} \right) + n \frac{-1}{1-p} \\ &= \left(\sum_{i=1}^m \hat{p}_i^{(n)} \right) \frac{1}{p(1-p)} - \frac{n}{1-p} \stackrel{\text{set}}{=} 0 \end{aligned}$$

$$\sum_{i=1}^m p_i^{(n)} = m \cdot p \Rightarrow p = \frac{\sum p_i^{(n)}}{m}$$

$$\hat{p}^{(n+1)} = \frac{\sum_{i=1}^m p_i^{(n)}}{m}$$

$$\hat{\mu}_1^{(n+1)} = \arg \max_{\mu_1} \sum_{i=1}^n (y_i - \mu_1)^2 \hat{p}_i^{(n)}$$

$$= \frac{\sum_{i=1}^n y_i \hat{p}_i^{(n)}}{\sum_{i=1}^n \hat{p}_i^{(n)}}$$

$$Q^{(n)}(\mu_1) = -\frac{1}{2} \sum_{i=1}^m (y_i - \mu_1)^2 \cdot \hat{p}_i^{(n)}$$

$$\frac{\partial Q^{(n)}(\mu_1)}{\partial \mu_1} \underset{\text{set}}{=} -\frac{1}{2} \sum_{i=1}^m 2 \cdot (\mu_1 - y_i) \cdot \hat{p}_i^{(n)} \underset{=}{=} 0$$

$$\hat{\mu}_0^{(n+1)} = \frac{\sum_{i=1}^m y_i (1 - \hat{p}_i^{(n)})}{\sum_{i=1}^m (1 - \hat{p}_i^{(n)})}$$

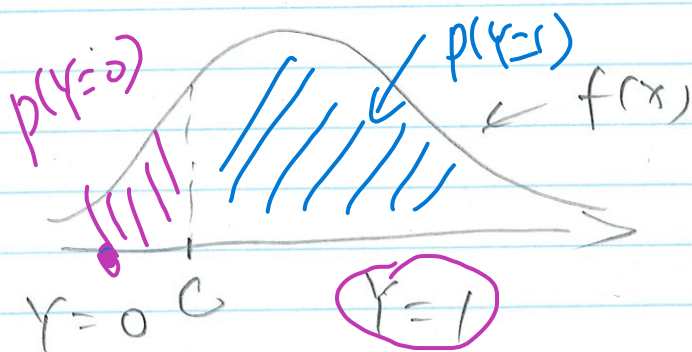
obtain $\hat{\Theta}^{(n+1)} = (\hat{p}^{(n+1)}, \hat{\mu}_1^{(n+1)}, \hat{\mu}_0^{(n+1)})$

Go to E-step to update $\hat{p}_i^{(n)}$

Truncated distribution

suppose $X \sim f(x)$ (e.g. normal, poisson)

let $Y = I(X > c)$, an indicator of $X > c$



$$Y = \begin{cases} 1, & X > c \\ 0, & X \leq c \end{cases}$$

Joint distribution of (X, Y) :

$$f_{XY}(x, y) = f(x) \cdot f_{Y|X}(y|x)$$

where the conditional distribution $Y|X$ is

$$f_{Y|X}(y|x) = \begin{cases} 1, & \text{if } y=1, X > c \\ 1, & \text{if } y=0, X \leq c \\ 0, & \text{if } y=0, X > c \\ 0, & \text{if } y=1, X \leq c \end{cases}$$

$$= [I(X > c)]^y [1 - I(X > c)]^{1-y}$$

Note: $0^0 = 1$

Marginal distribution of Y :

$$f_Y(y) = \int_{-\infty}^{+\infty} I(X > c)^y (1 - I(X > c))^{1-y} f(x) dx$$

$$f_Y(1) = \int_{-\infty}^{+\infty} I(x > c) f(x) dx$$

$$= \int_c^{+\infty} f(x) dx = P(X > c)$$

$$f_Y(0) = \int_{-\infty}^c f(x) dx = P(X \leq c)$$

$$Y \sim \text{Bern}(P(X > c))$$

Conditional distribution of $X | Y$:

$$f_{X|Y}(x | Y=1) = \frac{f_{XY}(x, Y=1)}{f_Y(1)}$$

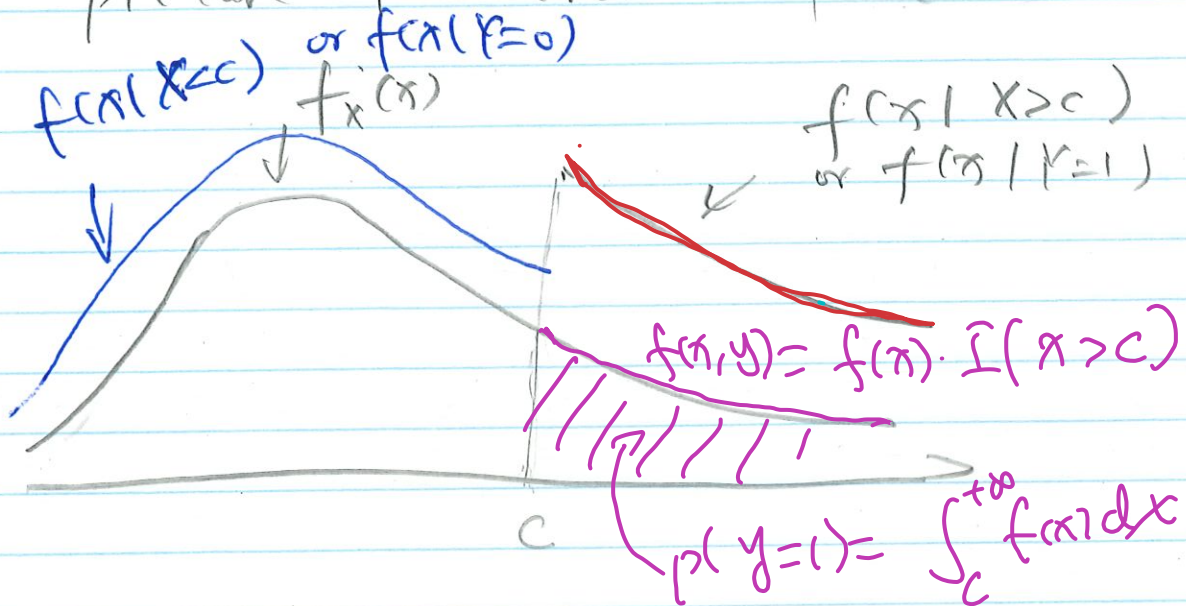
$$= \frac{f(x) \cdot I(x > c)}{P(X > c)}$$

$$= \begin{cases} \frac{f(x)}{\int_c^{+\infty} f(x) dx}, & \text{if } x > c \\ 0, & \text{o.w.} \end{cases}$$

Similarly,

$$f_{X|Y}(x | Y=0) = \frac{f(x)}{\int_{-\infty}^c f(x) dx} I(x \leq c)$$

A picture for truncated distribution



$x > c$: censored data

EM for censored poisson data

observed data

Form 1: $y_1, \dots, y_m, y_{m+1} < 2, \dots, y_n < 2$

Form 2: $y_1, \dots, y_m, x_{m+1} = 1, \dots, x_n = 1$ Form 3

where $x_i = I(y_i < 2)$

$y_i: y_1, \dots, y_m, 2, 2, \dots, 2$
 $\delta_i: 0, \dots, 0, 1, 1, \dots, 1$

Complete likelihood:

$f(y_1, \dots, y_m, y_{m+1}, \dots, y_n, x_{m+1}=1, \dots, x_n=1)$

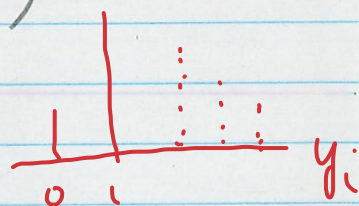
(Annotations: red circles around $y_1, \dots, y_m$ and $x_{m+1}=1, \dots, x_n=1$; purple circles around $y_{m+1}, \dots, y_n$; red arrows pointing to $y_{m+1}, \dots, y_n$ with label "missing"; red arrows pointing to $x_{m+1}=1, \dots, x_n=1$ with label "observed")

$$= \prod_{i=1}^m f^{\text{pois}}(y_i | \lambda) \cdot \prod_{i=m+1}^n f^{\text{pois}}(y_i | \lambda) \cdot \prod_{i=m+1}^n I(y_i < 2, x_i = 1)$$

Observed likelihood:

$$f(y_1, \dots, y_m, x_{m+1}=1, \dots, x_n=1)$$

$$= \prod_{i=1}^m f^{\text{pois}}(y_i | \lambda) \cdot \prod_{i=m+1}^n P(y_i < 2 | \lambda)$$



where $P(y_i < 2 | \lambda) = \sum_{y_i=0}^1 f^{\text{pois}}(y_i | \lambda)$

Let $Y = (y_1, \dots, y_m, y_{m+1}, \dots, y_n)$ ← missing

$X = (x_{m+1}, \dots, x_n)$

EM algorithm

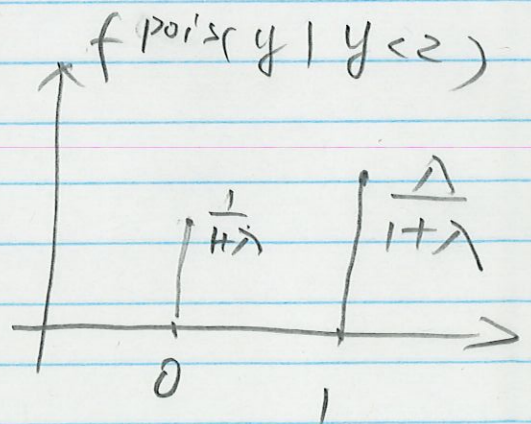
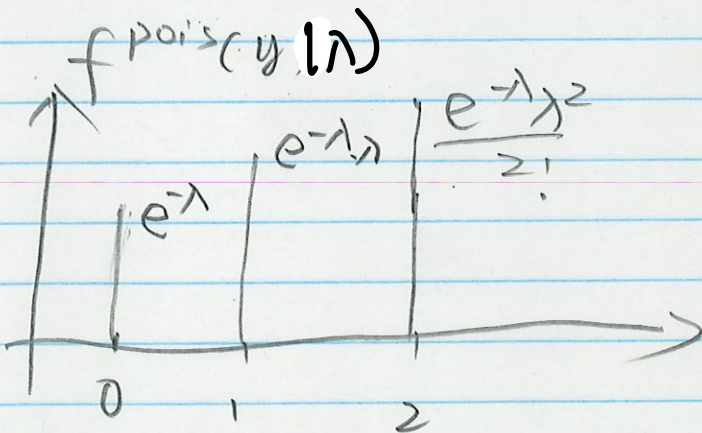
Given a previous estimate $\lambda^{(t)}$:

E-step: Compute $E_{Y|\lambda^{(t)}, X}(\log(f(Y, X|\lambda)))$

$$\begin{aligned}
 \ell^c(\lambda | Y, X) &= \sum_{i=1}^m \log\left(\frac{e^{-\lambda} \lambda^{y_i}}{y_i!}\right) + \sum_{i=m+1}^n \log\left(\frac{e^{-\lambda} \lambda^{y_i}}{y_i!}\right) + \text{missing} \\
 &= -n\lambda + (\log \lambda) \sum_{i=1}^m y_i + (\log \lambda) \underbrace{\sum_{i=m+1}^n y_i}_{\text{missing}} + \sum_{i=1}^n \log(y_i!)
 \end{aligned}$$

$$f(y_i | x_i=1, \lambda^{(t)}) = \begin{cases} \frac{1}{1 + \lambda^{(t)}} & y_i = 0 \\ \frac{\lambda^{(t)}}{1 + \lambda^{(t)}} & y_i = 1 \end{cases}$$

$\Downarrow$
 $y_i < 2$



$$E_{\lambda^{(t)}}(y_i | x_i=1) = \frac{\lambda^{(t)}}{1 + \lambda^{(t)}} \equiv \hat{y}_i^{(t)}$$

we don't need $\hat{y}_i^{(t)}$ to find $E_{\lambda^{(t)}}(\log(y_i!)) | x_i=1$!

E M is not just to replace y_i with $E(y_i | \lambda^{(t-1)})$

$$Q(\lambda | \lambda^{(t)})$$

$$= E_{\lambda^{(m)}} \ell^c(\lambda | Y, X)$$

$$= -n \lambda + \log \lambda \sum_{i=1}^m y_i + \log \lambda \sum_{i=m+1}^n E_{\lambda^{(t)}}(y_i | x_i) + C^{(t)}$$

$$= -n \lambda + \log \lambda \sum_{i=1}^m y_i + \log \lambda \sum_{i=m+1}^n \hat{y}_i^{(t)} + C^{(t)}$$

↑
unrelated
to λ

M-step:

$$\lambda^{(t+1)} = \underset{\lambda}{\operatorname{argmax}} Q(\lambda | \lambda^{(t)})$$

$$= \frac{\sum_{i=1}^m y_i + \sum_{i=m+1}^n \hat{y}_i^{(t)}}{n}$$

$$= \frac{m \bar{y}_{1:m} + (n-m) \frac{\lambda^{(t)}}{1 + \lambda^{(t)}}}{n}$$

Derivation of M-step, $C = n - m$

$$\frac{\partial Q}{\partial \lambda} = -n + \frac{1}{\lambda} \sum_{i=1}^m y_i + \frac{1}{\lambda} \sum_{i=m+1}^n y_i^{(t)} + 0 \stackrel{\text{set}}{=} 0$$

$$\Rightarrow \lambda = \frac{(\sum y_i + \sum y_i^{(t)})}{n}$$

Ascending Property of EM algorithm

⇒ EM algorithm

Observed y

Hidden z

⇒ Completed likelihood

$$\begin{aligned} L^c(\theta; y, z) &= P(y, z | \theta) \\ &= p(z | \theta) \cdot p(y | z, \theta) \end{aligned}$$

Note: Let $p(y, z | \theta) = h(y, z | \theta)$

$$g(y | \theta) = \int h(y, z | \theta) dz$$

⇒ observed likelihood:

$$L^o(\theta; y) = g(y | \theta)$$

Note: Let $T(z | \theta, y) = \frac{h(y, z | \theta)}{g(y | \theta)}$

⇒ EM - algorithm:

Given $\hat{\theta}^{(0)}$ repeat

E - step

$$Q(\theta | \hat{\theta}^{(m)}) = E_{z | y, \hat{\theta}^{(m)}} [\log(h(y, z | \theta))]]$$

$$= E_{\hat{\theta}^{(m)}} [\log(h(y, z | \theta))]]$$

M-step:

$$\hat{\theta}^{(m+1)} = \underset{\theta}{\operatorname{argmax}} Q(\theta | \hat{\theta}^{(m)})$$

⇒ Theorem 1: (Ascending of EM)

$$\begin{aligned} L^o(\hat{\theta}^{(m+1)} | y) &= q(y | \hat{\theta}^{(m+1)}) \\ &\geq L^o(\hat{\theta}^{(m)} | y) = q(y | \hat{\theta}^{(m)}) \end{aligned}$$

An Important Equation

$$\log L^o(\theta | y) = E_{z|\theta, y}(\log(h(y, z | \theta)))$$

$$= E_{z|\theta, y}(\log(K(z | \theta, y)))$$

(for any fixed θ)

$$Q(\theta | \theta^o) \leftarrow Q(\theta | \hat{\theta}^{(m)}) \text{ when } \hat{\theta}^{(m)} \rightarrow \theta^o$$

proof of the Theorem.

$$Q(\hat{\theta}^{(m+1)} | \hat{\theta}^{(m)}) \geq Q(\hat{\theta}^{(m)} | \hat{\theta}^{(m)})$$

By the identity

$$\begin{aligned} \log L^c(\hat{\theta}^{(m+1)} | y) &= E_{z | \hat{\theta}^{(m)}, y} [\log L^c(\hat{\theta}^{(m+1)} | y, z)] \\ &\quad - E_{z | \hat{\theta}^{(m)}, y} (\log K(z | \hat{\theta}^{(m+1)}, y)) \end{aligned}$$

$$\begin{aligned} \log L^c(\hat{\theta}^{(m)} | y) &= E_{z | \hat{\theta}^{(m)}} [\log L^c(\hat{\theta}^{(m)} | y, z)] \\ &\quad - E_{z | \hat{\theta}^{(m)}} [\log K(\hat{\theta}^{(m)} | y, z)] \end{aligned}$$

It suffices to show that

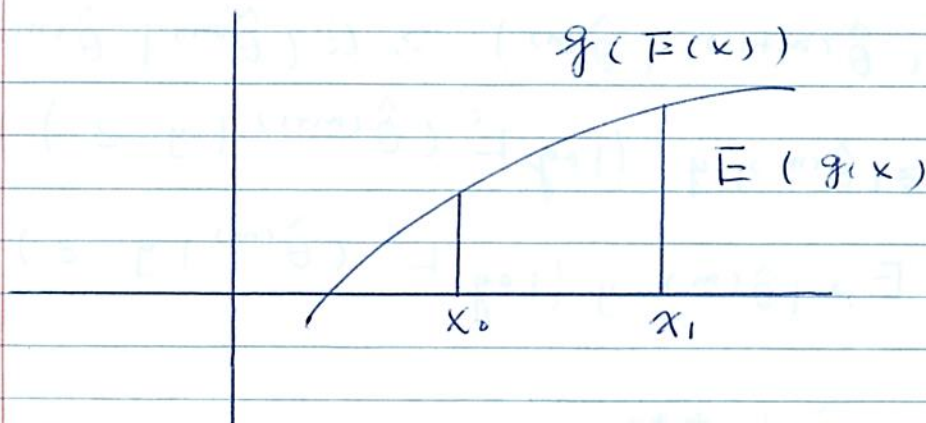
$$E_{z|\hat{\theta}^{(m)}, y} [\log K(z|y, \hat{\theta}^{(m+1)})]$$

$$\leq E_{z|\hat{\theta}^{(m)}, y} [\log K(z|y, \hat{\theta}^{(m)})]$$

$$\Leftrightarrow E_{z|\hat{\theta}^{(m)}, y} \left[\log \frac{K(z|y, \hat{\theta}^{(m+1)})}{K(z|y, \hat{\theta}^{(m)})} \right] \leq 0$$

By Jensen's Inequality

if $g(x)$ is a concave function



$$\Rightarrow E(g(x)) \leq g(E(x))$$

$$E_{z|\hat{\theta}^{(m)}, y} \left[\log \frac{K(z|y, \hat{\theta}^{(m+1)})}{K(z|y, \hat{\theta}^{(m)})} \right]$$

$$\leq \log \left[E_{z|\hat{\theta}^{(m)}, y} \left(\frac{K(z|y, \hat{\theta}^{(m+1)})}{K(z|y, \hat{\theta}^{(m)})} \right) \right]$$

$$= \log \int \frac{K(z|y, \hat{\theta}^{(m+1)})}{K(z|y, \hat{\theta}^{(m)})} \cdot K(z|\hat{\theta}^{(m)}, y) dz$$

$$= \log(1) = 0$$

Viewing EM as alternating optimization

Represented by a pdf or pmf $P(y)$,
as follows:

$$F(\underbrace{\tilde{P}(\cdot)}_{\substack{\text{unknown} \\ \text{distribution} \\ \text{about } y}}, \underbrace{\theta}_{\theta}) = \int \tilde{P}(y) \log(P(x, y | \theta)) dy \\ - \int \tilde{P}(y) \log(\tilde{P}(y)) dy$$

(1) when y is discrete

the integral in the above equation is
replaced by summation

(2) We will show that

$F(\tilde{P}(\cdot), \theta)$ $\swarrow$ E - step is equivalent to Maximizing
Fixed F with θ fixed and

$\swarrow$ M - step is equivalent to Maximizing
 $F(\tilde{P}(\cdot), \theta)$ $\swarrow$ F with $\tilde{P}(\cdot)$ fixed
Fixed

First: we can rewrite F in another way:

$$F(\hat{P}(\cdot), \theta) \\ = \int \tilde{P}(y) \log(P(x, y | \theta)) dy - \int \tilde{P}(y) \log(P(y|x, \theta)) dy$$

$$= \int \tilde{P}(y) \log(P(x|\theta)) dy - \int \tilde{P}(y) \log\left(\frac{\tilde{P}(y)}{P(y|x, \theta)}\right) dy$$

Note that: joint PDF of R.V. x, y

$$f_{x,y}(x, y) = f_{y|x}(y|x) f_x(x) //$$

$$= \log(P(x|\theta)) \int \tilde{P}(y) dy - \int \tilde{P}(y) \log\left(\frac{\tilde{P}(y)}{P(y|x, \theta)}\right) dy$$

$$= \log P(x|\theta) - \int \tilde{P}(y) \log\left(\frac{\tilde{P}(y)}{P(y|x, \theta)}\right) dy$$

$$= L(\theta, x) - KL(\hat{P}(\cdot) \| P_{y|x}(\cdot|x, \theta))$$

where, $L(\theta, x) = \log P(x|\theta)$ is the log likelihood function of θ based on the observed data only

$$KL(P(\cdot) \| Q(\cdot)) = \int \tilde{P}(y) \log \left(\frac{\tilde{P}(y)}{P(y|x, \theta)} \right) dy$$

is called the Kullback-Leibler divergence between two distributions $P(\cdot)$ and $Q(\cdot)$

Second: we now maximize $F(\tilde{P}(\cdot), \theta)$, using

alternative procedure. Fixing θ at some value say $\theta^{(0)}$, the $\tilde{P}(\cdot)$ maximizes $F(\tilde{P}(\cdot), \theta^{(0)})$ is the $\tilde{P}(\cdot)$ minimizes $KL(\tilde{P}(\cdot) \| P(\cdot|x, \theta^{(0)}))$

Because the first term ($L(\theta, x)$) has nothing with $\tilde{P}(\cdot)$. This optimization will lead to choosing $P(\cdot|x, \theta^{(0)})$ as the $\tilde{P}(\cdot)$. Since it is well-known that $KL(P(\cdot) \| Q(\cdot))$ is minimized to 0 when $P=Q$, shown by the following inequality:

show that

$$KL(P_{(\cdot)} \| Q_{(\cdot)})$$

$$= \int P(y) \log(P(y)/Q(y)) dy$$

$$= - \int P(y) \log(Q(y)/P(y)) dy$$

$$\geq - \int P(y) \left(\frac{Q(y)}{P(y)} - 1 \right) dy$$

(Note that: since $\log x \leq x - 1$)

$$= - \int \left[P(y) \cdot \frac{Q(y)}{P(y)} - P(y) \right] dy$$

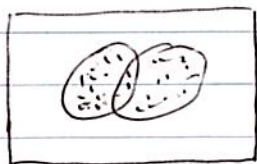
$$= - \int [Q(y) - P(y)] dy$$

$$= 0 \quad (\text{when } P(\cdot) = Q(\cdot))$$

This step is equivalent to E-step

=> Some advanced examples of E.M.

1) Model-based clustering



$$y_i | z_i \sim N(\mu, \Sigma), \quad z_i \sim \text{Cathy}(\cdot)$$

2) Hidden Markov model

$$\begin{array}{ccccccc} y_1, & y_2, & \dots & y_n \\ \uparrow & \uparrow & & \uparrow \\ h_1 & h_2 & \dots & h_n \end{array}$$

$$P(y_1, \dots, y_n | h_1, \dots, h_n, \theta)$$

$$= \prod_{i=1}^n P(y_i | h_i, \theta)$$

3) Factor Analysis Model.

$$y = T'x + \varepsilon$$

$$y_i = p_i x_i + \varepsilon_i$$

↓

observed

↓

hidden

$$E\text{-step: } x_i | y_i, \hat{\theta}^{(n)}$$